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Quantum Computing as a Catalyst for Microgrid Management: Enhancing Decentralized Energy Systems Through Innovative Computational Techniques

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Abstract: This paper introduces a groundbreaking framework for optimizing microgrid operations using the Quantum Approximate Optimization Algorithm (QAOA). The increasing integration of decentralized energy systems, characterized by their reliance on renewable energy sources, presents unique challenges, including the stochastic nature of energy supply-and-demand management. Our study leverages quantum computing to enhance the operational efficiency and resilience of microgrids, transcending the limitations of traditional computational methods. The proposed QAOA-based model formulates the microgrid scheduling problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem, suitable for quantum computation. This approach not only accommodates complex operational constraints—such as energy conservation, peak load management, and cost efficiency—but also dynamically adapts to the variability inherent in renewable energy sources. By encoding these constraints into a quantum-friendly Hamiltonian, QAOA facilitates a parallel exploration of multiple potential solutions, enhancing the probability of reaching an optimal solution within a feasible time frame. We validate our model through a comprehensive simulation using real-world data from a microgrid equipped with photovoltaic systems, wind turbines, and energy storage units. The results demonstrate that QAOA outperforms conventional optimization techniques in terms of cost reduction, energy efficiency, and system reliability. Furthermore, our study explores the scalability of quantum algorithms in energy systems, providing insights into their potential to handle larger, more complex grid architectures as quantum technology advances. This research not only underscores the viability of quantum algorithms in real-world applications but also sets a precedent for future studies on the integration of quantum computing into energy management systems, paving the way for more sustainable, efficient, and resilient energy infrastructures.

Keywords: quantum computing; microgrid optimization; decentralized energy systems; quantum approximate optimization algorithm (QAOA); sustainable energy management; energy system resilience

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1. Introduction

The increasing integration of renewable energy sources and the widespread deployment of microgrids are reshaping modern power systems, particularly in rural and remote areas where centralized electricity infrastructure remains limited or unreliable [1]. Rural

microgrids, characterized by decentralized control and a high penetration of distributed energy resources (DERs), play a critical role in enhancing energy accessibility, sustainability, and resilience in isolated communities [2,3]. These microgrids rely on a mix of renewable generation such as solar photovoltaic (PV) systems and wind turbines, coupled with energy storage systems (ESSs) to maintain a stable and reliable power supply [4]. However, the intermittent and stochastic nature of renewable energy sources, coupled with seasonal fluctuations in demand, introduces significant operational challenges that require sophisticated optimization techniques to ensure efficient resource allocation [5,6]. To address these challenges, this study considers a hybrid AC/DC microgrid architecture with islanding capability, which reflects the operational characteristics of many rural and remote community energy systems. In this configuration, alternating current (AC) lines are used to support traditional residential and commercial loads, while direct current (DC) links are dedicated to the efficient integration of renewable sources such as photovoltaic (PV) systems and battery energy storage systems (BESSs). This dual-layered approach not only enhances power conversion efficiency by minimizing unnecessary AC-DC conversions but also supports flexible infrastructure deployment in areas with heterogeneous energy demands. Moreover, the microgrid is equipped with islanding functionality, allowing it to seamlessly disconnect from the main utility grid and operate autonomously during outages or disturbances. This ensures supply continuity for critical loads and bolsters energy security in regions where the central grid is unreliable or inaccessible. The hybrid AC/DC and islanded design presents unique operational constraints and coordination challenges, which motivate the need for advanced quantum-enhanced scheduling and optimization strategies. This paper specifically addresses these challenges within the context of such a microgrid framework.

Traditional methods for optimizing rural microgrid operations often rely on deterministic or heuristic approaches, which struggle to handle the high-dimensional, combinatorial nature of energy dispatch decisions, particularly in the presence of multiple uncertainties.

As power systems move toward a more decentralized and intelligent architecture, there is an increasing need for advanced computational techniques that can efficiently solve complex optimization problems in real time. Recent advancements in quantum computing have opened up new possibilities for tackling high-dimensional optimization challenges that were previously computationally intractable for classical solvers. The Quantum Approximate Optimization Algorithm (QAOA) is a leading quantum-inspired optimization method that has demonstrated remarkable efficiency in solving combinatorial problems with discrete decision variables [7]. By leveraging quantum superposition and entanglement, QAOA offers an unprecedented capability to explore multiple solution spaces simultaneously, significantly accelerating the convergence toward near-optimal energy dispatch strategies [8]. This paper presents a novel QAOA-based optimization framework for resource allocation in distributed rural microgrids, designed to handle the dynamic interplay of renewable generation, storage utilization, and demand-side flexibility while ensuring cost-effectiveness, reliability, and decarbonization.

Figure 1 illustrates the schematic overview of the proposed QAOA-based quantum–classical optimization framework for microgrid scheduling. The framework is composed of three interconnected modules: classical pre-processing, quantum optimization, and post-processing analysis. At the lower left, the system begins with classical inputs including microgrid topology, renewable generation and load data, and operational constraints. These inputs are encoded into a QUBO (Quadratic Unconstrained Binary Optimization) formulation, which serves as the input structure compatible with QAOA. The QUBO problem is then passed to the QAOA-based quantum layer, shown at the top of the figure,

where the cost Hamiltonian is constructed, variational parameters are initialized and tuned, and the quantum–classical loop is executed to iteratively minimize the objective function.

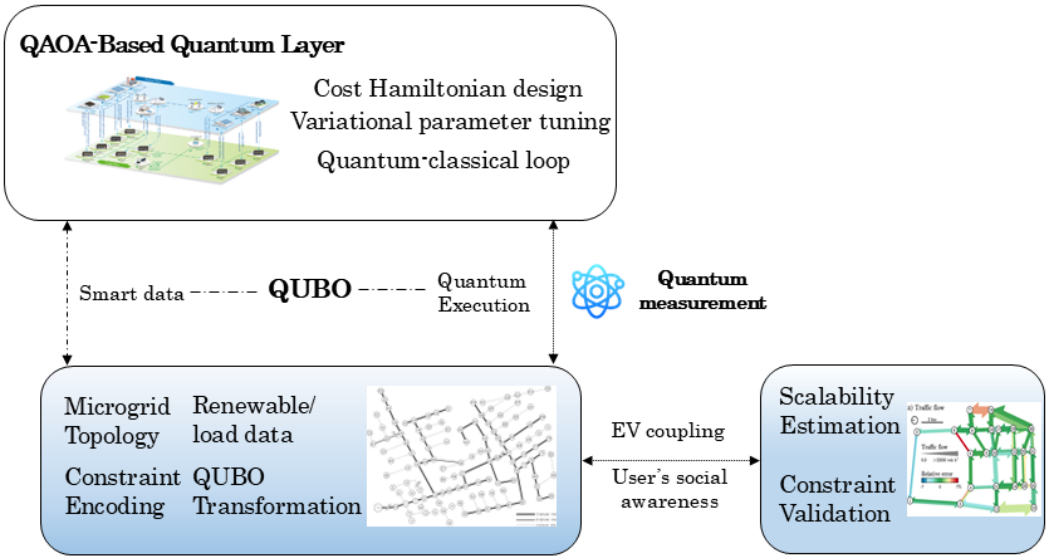

Figure 1. Quantum–classical architecture for scalable microgrid optimization.

Following quantum execution, the system proceeds to post-processing, as shown on the right side of the figure. Here, quantum measurement results are interpreted and validated against operational constraints. Additionally, the scalability of the proposed approach is evaluated to ensure its viability under increased system complexity. The entire process embodies the synergy between classical problem modeling and quantum computational acceleration, paving the way for scalable, real-time, and resilient microgrid energy management.

The proposed framework formulates the microgrid resource scheduling problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem, which is inherently compatible with QAOA's quantum variational approach. Unlike conventional linear programming or mixed-integer optimization techniques, which scale poorly with increasing system complexity, the QUBO-based formulation enables efficient encoding of energy dispatch constraints into a quantum-friendly cost function. The optimization problem integrates multiple objectives, including operational cost minimization, energy self-sufficiency maximization, system reliability enhancement, and robust scheduling under uncertainty. These objectives are embedded into a hybrid quantum–classical architecture, where the quantum processor iteratively evolves variational quantum states to approximate the optimal microgrid dispatch schedule, while a classical optimizer refines the quantum parameters to enhance solution accuracy. This hybrid strategy ensures that the computational advantages of quantum optimization are effectively utilized while maintaining compatibility with practical microgrid operations. The contributions of this paper are fourfold.

First, it introduces a novel quantum-inspired optimization framework for rural microgrid resource allocation, leveraging QAOA to handle the combinatorial complexity of energy scheduling.

Second, it formulates microgrid scheduling as a QUBO problem, ensuring seamless compatibility with quantum variational methods while incorporating key operational constraints such as power balance, storage utilization, and grid interaction.

Third, it develops a hybrid quantum–classical optimization architecture, where quantum state evolution is iteratively refined using classical machine learning techniques to enhance solution quality.

Finally, it conducts an extensive performance evaluation using real-world microgrid data, demonstrating the superiority of QAOA-based scheduling in terms of both computational efficiency and economic viability.

2. Literature Review

Traditional methods for microgrid optimization have relied heavily on deterministic approaches, such as mixed-integer linear programming (MILP) and mixed-integer nonlinear programming (MINLP), which explicitly model power flow equations, operational constraints, and cost functions. These methods have been extensively applied to solve economic dispatch problems, generation scheduling, and storage management [9,10]. While deterministic models provide mathematically rigorous solutions, they suffer from scalability issues, particularly when dealing with the high-dimensional optimization landscape of microgrids with a large number of DERs and dynamic constraints. Additionally, deterministic models often require precise forecasting of renewable generation and load demand, which is inherently uncertain [11,12]. As a result, stochastic optimization has been introduced to account for variability in energy supply and demand. Stochastic programming methods, such as scenario-based approaches and chance-constrained optimization, have improved robustness by incorporating probabilistic constraints and uncertainty distributions [13]. However, these methods significantly increase computational complexity, as the number of scenarios grows exponentially with the problem size, making them impractical for real-time microgrid operations [14].

To overcome the computational burden of deterministic and stochastic approaches, metaheuristic algorithms, such as genetic algorithms (GAs), particle swarm optimization (PSO), and simulated annealing (SA), have been widely adopted for microgrid scheduling. These nature-inspired optimization techniques provide flexibility in handling complex, nonlinear, and multi-objective problems while requiring fewer assumptions about the problem structure [15]. For example, PSO has been effectively applied to optimize DER dispatch by iteratively refining solutions based on swarm intelligence, while GA has demonstrated effectiveness in optimizing microgrid resilience under extreme weather conditions [16,17]. Despite their advantages, metaheuristic methods suffer from convergence issues and require extensive parameter tuning, which limits their applicability for large-scale microgrid scheduling problems. Furthermore, these algorithms often struggle to find globally optimal solutions, particularly in scenarios with highly constrained energy resources, leading to suboptimal performance.

In recent years, reinforcement learning (RL) and deep learning-based approaches have gained traction in microgrid optimization, leveraging the ability of artificial intelligence (AI) to learn complex patterns from data and adapt to changing environments. Deep reinforcement learning (DRL), in particular, has been explored for demand response management, real-time energy trading, and optimal control of storage systems [18,19]. Techniques such as Deep Q-Networks (DQNs) and Proximal Policy Optimization (PPO) have been employed to develop adaptive energy scheduling strategies that dynamically adjust to fluctuations in renewable generation [20]. Although DRL offers a data-driven approach to optimization, its reliance on large training datasets and extensive computational resources presents significant challenges for real-time implementation in microgrids. Moreover, the black-box nature of neural networks raises concerns regarding interpretability and trustworthiness in critical power system applications. With the limitations of classical and AI-driven methods, quantum computing has emerged as a promising avenue for tackling high-dimensional optimization problems in energy systems. Quantum computing leverages quantum mechanical principles such as superposition, entanglement, and interference to process information in ways that classical computers cannot efficiently replicate [21].

Among various quantum algorithms, QAOA has gained significant attention for solving combinatorial optimization problems, including energy scheduling, logistics, and portfolio optimization. QAOA operates by encoding the optimization problem into a Hamiltonian representation and evolving quantum states to iteratively approximate the optimal solution. Unlike classical solvers that sequentially explore solution spaces, QAOA exploits quantum parallelism, enabling simultaneous evaluation of multiple potential solutions.

Several studies have begun exploring the application of QAOA in energy optimization, particularly in power system planning and grid operation [22]. Researchers have demonstrated that QAOA can efficiently solve unit commitment problems, optimize transmission network expansion, and enhance renewable energy forecasting. However, its application in microgrid scheduling remains largely unexplored [23]. While existing quantum-inspired optimization frameworks have shown potential, they often lack a structured methodology for integrating microgrid-specific constraints, such as energy balancing, voltage stability, and demand-side flexibility. Additionally, previous work on quantum computing in power systems has largely been theoretical, with limited real-world validation using empirical energy demand and generation data. This gap in the literature underscores the need for a comprehensive, application-driven QAOA framework tailored specifically for microgrid resource allocation.

3. Problem Formulation and the Proposed Method

In this section, we detail the problem formulation and elaborate on the proposed method for optimizing microgrid operations using the Quantum Approximate Optimization Algorithm (QAOA). Given the complexity and dynamic nature of microgrid systems, especially those integrating various renewable energy sources, a sophisticated approach is required to manage the myriad operational challenges effectively. These challenges encompass maintaining energy balance, minimizing operational costs, maximizing the use of renewable resources, and ensuring system reliability under varying environmental conditions.

Our approach utilizes a hybrid quantum–classical computational framework to address these challenges. We begin by defining the microgrid's operational constraints and objectives in a mathematical model suitable for quantum optimization. Subsequently, we introduce QAOA as the central mechanism of our optimization process, detailing its integration with classical computational techniques to enhance its efficiency and applicability. This section aims to provide a comprehensive understanding of how quantum computational advantages can be harnessed to solve complex optimization problems in energy systems, bridging the gap between theoretical quantum mechanics and practical energy management solutions.

$$\min_{\mathbf{\Phi},\mathbf{\Lambda},\mathbf{\Psi}} \sum_{t=1}^{T} \sum_{\iota\in\mathcal{N}} \left(\omega_{\iota,t}^{\text{gen}} \cdot \varphi_{\iota,t}^{\text{ren}} + \lambda_{\iota,t}^{\text{stor}} \cdot \psi_{\iota,t}^{\text{deg}} + \rho_{\iota,t}^{\text{main}} \cdot \chi_{\iota,t}^{\text{op}} + \beta_{\iota,t} \cdot \theta_{\iota,t} \right) + \sum_{t=1}^{T} \sum_{\kappa\in\mathcal{G}} \tau_{\kappa,t} \cdot \delta_{\kappa,t} \quad (1)$$

Minimizing the total operational cost in a rural microgrid is a complex task requiring consideration of multiple dynamic factors. This formulation integrates renewable energy generation costs, $\omega_{\iota,t}^{\text{gen}}$ weighted by the generated power $\varphi_{\iota,t}^{\text{ren}}$, storage degradation costs, $\lambda_{\iota,t}^{\text{stor}}$ linked with the discharged energy $\psi_{\iota,t}^{\text{deg}}$, and ongoing maintenance costs $\rho_{\iota,t}^{\text{main}}$ associated with the operational state $\chi_{\iota,t}^{\text{op}}$. Additional penalty terms $\beta_{\iota,t} \cdot \theta_{\iota,t}$ account for energy balance constraints. Furthermore, imported energy purchases from the main grid, $\tau_{\kappa,t} \cdot \delta_{\kappa,t}$, contribute to the cost function, ensuring cost-efficient microgrid operations.

In traditional microgrid scheduling models, cost coefficients such as renewable generation cost, storage degradation cost, and grid electricity price are often assumed to be

constant over time. However, this assumption fails to reflect the nonlinear nature of economic dynamics in practical microgrids. As rightly pointed out by the reviewer, real-world systems experience cost escalations driven by a combination of factors, including the exponential degradation of battery health, inflation-driven increases in electricity prices, and aging-induced maintenance overheads. To incorporate these realistic phenomena into our quantum optimization framework, we extend the cost function formulation by embedding time-dependent exponential cost coefficients. Specifically, the unit costs for generation, storage degradation, and energy import are modeled as follows:

$$\omega_{\iota,t}^{\text{gen}} = \omega_{\iota,0} \cdot e^{\alpha_{\text{gen}} \cdot t}, \quad \lambda_{\iota,t}^{\text{stor}} = \lambda_{\iota,0} \cdot e^{\alpha_{\text{stor}} \cdot t}, \quad \tau_{\kappa,t} = \tau_{\kappa,0} \cdot e^{\alpha_{\text{grid}} \cdot t} \tag{2}$$

Here, $\omega_{\iota,0}$, $\lambda_{\iota,0}$, and $\tau_{\kappa,0}$ denote the initial unit cost coefficients for renewable energy production, storage wear, and grid import, respectively. The exponential scaling factors α_{gen}, α_{stor}, and α_{grid} capture the time-varying nature of cost escalation and are determined from empirical observations. For instance, $\alpha_{\text{stor}} = 0.09$ per annum corresponds to a 10% annual increase in degradation-related maintenance cost, aligning with typical lithium-ion battery wear profiles. By substituting these into the original objective function Equation (1), we obtain a temporally nonstationary cost structure:

$$\min \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\omega_{\iota,0} e^{\alpha_{\text{gen}} t} \cdot \varphi_{\iota,t}^{\text{ren}} + \lambda_{\iota,0} e^{\alpha_{\text{stor}} t} \cdot \psi_{\iota,t}^{\text{deg}} + \rho_{\iota,t}^{\text{main}} \cdot \chi_{\iota,t}^{\text{op}} + \beta_{\iota,t} \cdot \theta_{\iota,t} \right) + \sum_{t=1}^{T} \sum_{\kappa \in \mathcal{G}} \tau_{\kappa,0} e^{\alpha_{\text{grid}} t} \cdot \delta_{\kappa,t} \tag{3}$$

This refined formulation allows the QAOA-based optimization to reflect the growing economic cost of prolonged operation, especially in off-grid or storage-dominant periods. Furthermore, these exponential cost functions are directly embedded into the quantum Hamiltonian $\mathcal{H}_{\text{cost}}$ (Equation (29)),ensuring that the quantum system explores optimal configurations that not only satisfy physical constraints but also adapt to rising operational costs over time. The use of exponential modeling enhances the realism of our simulations and aligns the energy scheduling decisions with lifecycle-aware and inflation-sensitive strategies. This provides a stronger basis for long-term microgrid sustainability and robustness against external market volatility.

$$\max_{\Gamma} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\frac{\varphi_{\iota,t}^{\text{ren}} + \zeta_{\iota,t}^{\text{stor,dis}} - \delta_{\iota,t}^{\text{imp}}}{\psi_{\iota,t}^{\text{dem}}} \right) \cdot \gamma_{\iota,t}^{\text{self}} \tag{4}$$

Enhancing energy self-sufficiency is essential for minimizing dependency on external sources and ensuring microgrid resilience. This equation formulates self-sufficiency as the fraction of locally generated power $\varphi_{\iota,t}^{\text{ren}}$ and storage discharge $\zeta_{\iota,t}^{\text{stor,dis}}$, minus the imported energy $\delta_{\iota,t}^{\text{imp}}$, over the total demand $\psi_{\iota,t}^{\text{dem}}$. The term $\gamma_{\iota,t}^{\text{self}}$ represents an adaptive weight that adjusts based on seasonality and local grid conditions. A high self-sufficiency ratio signifies a more independent and robust microgrid system.

$$\max_{\Omega} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\xi_{\iota,t} \cdot \left| \varphi_{\iota,t}^{\text{ren}} + \zeta_{\iota,t}^{\text{stor,dis}} - \psi_{\iota,t}^{\text{dem}} \right| + \eta_{\iota,t} \cdot \left| \delta_{\iota,t}^{\text{imp}} - \psi_{\iota,t}^{\text{res}} \right| \right) \tag{5}$$

Reliability in a rural microgrid requires supply–demand balance and sufficient reserves. This equation seeks to maximize reliability by penalizing deviations between generated energy $\varphi_{\iota,t}^{\text{ren}}$, storage discharge $\zeta_{\iota,t}^{\text{stor,dis}}$, and demand $\psi_{\iota,t}^{\text{dem}}$, weighted by $\xi_{\iota,t}$. The second term ensures reserve sufficiency by controlling the difference between imported energy $\delta_{\iota,t}^{\text{imp}}$ and reserved energy $\psi_{\iota,t}^{\text{res}}$, modulated by $\eta_{\iota,t}$. This formulation enforces robust reliability in multi-seasonal energy operation.

$$\min_{\boldsymbol{\Theta}} \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\sum_{p=1}^{P} \alpha_p \cos\left(\boldsymbol{\theta}_{\iota,t}^{p}\right) + \sum_{q=1}^{Q} \beta_q \sin\left(\boldsymbol{\lambda}_{\iota,t}^{q}\right) \right) \tag{6}$$

Quantum optimization plays a pivotal role in refining energy allocation in the microgrid, and this equation encapsulates the QAOA-based variational cost function. It leverages quantum state parameterization through $\boldsymbol{\theta}_{\iota,t}^{p}$ and $\boldsymbol{\lambda}_{\iota,t}^{q}$, representing decision variables encoded as quantum states. The summation over cosine and sine terms, weighted by α_p and β_q, respectively, allows the quantum system to explore diverse configurations and converge toward an optimal solution via quantum measurement. This formulation enables quantum-enhanced energy scheduling and resource allocation.

$$\begin{aligned} &\sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\zeta_{\iota,t}^{\text{stor,chg}} - \zeta_{\iota,t}^{\text{stor,dis}} \right) + \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\varphi_{\iota,t}^{\text{ren}} + \delta_{\iota,t}^{\text{imp}} - \psi_{\iota,t}^{\text{dem}} - \psi_{\iota,t}^{\text{res}} \right) \\ &+ \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left(\eta_{\iota,t}^{\text{loss}} + \xi_{\iota,t}^{\text{curt}} \right) = 0 \end{aligned} \tag{7}$$

Maintaining power balance is fundamental in microgrid operation to ensure stable system performance. This equation enforces that the total energy charged into storage $\zeta_{\iota,t}^{\text{stor,chg}}$ minus the energy discharged $\zeta_{\iota,t}^{\text{stor,dis}}$ must align with the sum of renewable generation $\varphi_{\iota,t}^{\text{ren}}$, imported energy $\delta_{\iota,t}^{\text{imp}}$, and demand $\psi_{\iota,t}^{\text{dem}}$ while also accounting for the reserved power $\psi_{\iota,t}^{\text{res}}$. Additionally, line losses $\eta_{\iota,t}^{\text{loss}}$ and curtailed energy $\xi_{\iota,t}^{\text{curt}}$ must be included to ensure accurate power conservation across the microgrid. This formulation provides a holistic representation of the energy dynamics in the system.

$$\begin{aligned} &\zeta_{\iota,t}^{\text{stor,chg}} \leq \lambda_{\iota}^{\text{stor,max}} \cdot \sigma_{\iota,t}^{\text{stor,chg}}, \quad \zeta_{\iota,t}^{\text{stor,dis}} \leq \lambda_{\iota}^{\text{stor,max}} \cdot \sigma_{\iota,t}^{\text{stor,dis}} \\ &\text{s.t.} \quad \sigma_{\iota,t}^{\text{stor,chg}} + \sigma_{\iota,t}^{\text{stor,dis}} \leq 1, \quad \forall t, \iota \end{aligned} \tag{8}$$

Storage system operations are constrained by their respective charge and discharge limits, ensuring that at any given time, storage cannot charge and discharge simultaneously. Here, $\zeta_{\iota,t}^{\text{stor,chg}}$ and $\zeta_{\iota,t}^{\text{stor,dis}}$ are capped by the storage capacity limit $\lambda_{\iota}^{\text{stor,max}}$, with binary decision variables $\sigma_{\iota,t}^{\text{stor,chg}}$ and $\sigma_{\iota,t}^{\text{stor,dis}}$ ensuring mutual exclusivity. This guarantees that the microgrid storage functions optimally under practical constraints.

$$\psi_{\iota,t}^{\text{res}} \geq \eta_{\iota,t}^{\text{min,res}} + \sum_{k \in \mathcal{K}} \tau_{k}^{\text{risk}} \cdot \chi_{\iota,k,t}^{\text{load,var}} \tag{9}$$

Ensuring adequate reserve power is crucial for handling uncertainties in renewable generation and demand fluctuations. This constraint mandates that the reserved power $\psi_{\iota,t}^{\text{res}}$ must always meet or exceed a minimum threshold $\eta_{\iota,t}^{\text{min,res}}$, which accounts for contingency requirements. Additionally, an adaptive risk-aware term is included, where τ_{k}^{risk} quantifies the uncertainty impact from load variations $\chi_{\iota,k,t}^{\text{load,var}}$, ensuring robustness against unforeseen demand spikes.

$$\sum_{\iota \in \mathcal{N}} \varphi_{\iota,t}^{\text{ren}} \leq \sum_{\iota \in \mathcal{N}} \mu_{\iota,t}^{\text{ren,max}} \cdot \left(1 - \xi_{\iota,t}^{\text{curt}}\right) \tag{10}$$

Renewable energy generation is inherently constrained by environmental factors and technological limitations. This equation ensures that total renewable generation $\varphi_{\iota,t}^{\text{ren}}$ does not exceed the maximum potential $\mu_{\iota,t}^{\text{ren,max}}$ at any given time. The term $\left(1 - \xi_{\iota,t}^{\text{curt}}\right)$ accounts for potential curtailment, allowing the system to dynamically adjust generation levels while maintaining grid stability.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\varphi_{\iota,t}^{\text{ren}}+\zeta_{\iota,t}^{\text{stor,dis}}-\delta_{\iota,t}^{\text{imp}}-\psi_{\iota,t}^{\text{dem}}\right)\cdot\gamma_{\iota,t}^{\text{net}}=0 \tag{11}$$

This equation enforces the energy balance condition in the net energy flow dynamics. The total energy generated and discharged from storage must align with demand and imported energy, ensuring the system does not accumulate excess power. The coefficient $\gamma_{\iota,t}^{\text{net}}$ allows adjustments based on energy balancing needs, adapting to microgrid operational conditions.

$$\delta_{\iota,t}^{\text{imp}}\leq\mu_{\iota,t}^{\text{imp,max}}\cdot\sigma_{\iota,t}^{\text{grid,imp}},\quad\forall\iota,t \tag{12}$$

Energy imports from the external grid must be limited to prevent excessive reliance on external power sources. This constraint ensures that imported power $\delta_{\iota,t}^{\text{imp}}$ does not exceed the allowed capacity $\mu_{\iota,t}^{\text{imp,max}}$, with the binary decision variable $\sigma_{\iota,t}^{\text{grid,imp}}$ governing whether import is allowed at time t.

$$\sum_{\iota\in\mathcal{N}}\left(\varphi_{\iota,t}^{\text{ren}}+\zeta_{\iota,t}^{\text{stor,dis}}\right)-\sum_{\iota\in\mathcal{N}}\left(\psi_{\iota,t}^{\text{dem}}+\delta_{\iota,t}^{\text{imp}}\right)\leq\epsilon_{\text{stability}},\quad\forall t \tag{13}$$

Maintaining grid stability requires that the absolute power imbalance remains within an acceptable range. This constraint ensures that the sum of local generation and storage discharge does not significantly exceed total demand and imports, with a margin of deviation $\epsilon_{\text{stability}}$ allowed to accommodate minor fluctuations.

$$\sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\left(\varphi_{\iota,t}^{\text{ren}}+\zeta_{\iota,t}^{\text{stor,dis}}-\delta_{\iota,t}^{\text{imp}}-\psi_{\iota,t}^{\text{dem}}\right)^{2}\cdot\gamma_{\iota,t}^{\text{penalty}} \tag{14}$$

Minimizing energy imbalances over the entire planning horizon helps maintain an efficient and resilient microgrid. This quadratic penalty function ensures that deviations from perfect energy balancing are penalized, discouraging inefficient resource allocation and promoting optimal scheduling.

$$\begin{aligned}&\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\varphi_{\iota,t}^{\text{ren}}+\zeta_{\iota,t}^{\text{stor,dis}}-\delta_{\iota,t}^{\text{imp}}-\psi_{\iota,t}^{\text{dem}}-\psi_{\iota,t}^{\text{res}}\right)\cdot\gamma_{\iota,t}^{\text{bal}}\\&+\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\eta_{\iota,t}^{\text{loss}}+\xi_{\iota,t}^{\text{curt}}\right)\cdot\beta_{\iota,t}^{\text{var}}\leq\epsilon_{\text{stability}}\end{aligned} \tag{15}$$

Ensuring that energy balance is maintained at all times is critical for system stability. This equation extends the energy balance formulation by incorporating weighted coefficients $\gamma_{\iota,t}^{\text{bal}}$ that adaptively scale the balance constraint based on real-time operational requirements. Additionally, losses $\eta_{\iota,t}^{\text{loss}}$ and curtailed energy $\xi_{\iota,t}^{\text{curt}}$ are integrated with dynamic weight factors $\beta_{\iota,t}^{\text{var}}$ to ensure robustness against variability. The right-hand side allows a small deviation $\epsilon_{\text{stability}}$ to accommodate short-term fluctuations without compromising the stability of the system.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\zeta_{\iota,t}^{\text{stor,chg}}\cdot\alpha_{\iota,t}^{\text{eff,chg}}-\zeta_{\iota,t}^{\text{stor,dis}}\cdot\alpha_{\iota,t}^{\text{eff,dis}}\right)\leq\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\lambda_{\iota}^{\text{stor,max}} \tag{16}$$

This equation governs the efficiency-adjusted energy flow in storage systems. It ensures that the total energy charged into storage, scaled by charging efficiency $\alpha_{\iota,t}^{\text{eff,chg}}$, minus the energy discharged, adjusted by discharging efficiency $\alpha_{\iota,t}^{\text{eff,dis}}$, remains within the physical storage capacity $\lambda_{\iota}^{\text{stor,max}}$. This formulation captures realistic losses in energy cycling processes.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\zeta_{\iota,t}^{\text{stor,dis}}-\zeta_{\iota,t}^{\text{stor,chg}}\right)+\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\delta_{\iota,t}^{\text{imp}}-\psi_{\iota,t}^{\text{dem}}\right)\leq\tau^{\text{grid,max}} \tag{17}$$

Grid constraints must be strictly observed to prevent overloading. This constraint ensures that the difference between discharged and charged storage power, along with the net imported energy after subtracting demand, remains within the grid's maximum allowable power exchange limit $\tau^{\text{grid,max}}$. This formulation prevents operational violations that could disrupt microgrid stability.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\varphi_{\iota,t}^{\text{ren}}-\xi_{\iota,t}^{\text{curt}}\right)\leq\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\lambda_{\iota,t}^{\text{ren,max}} \tag{18}$$

Renewable generation limits must be strictly enforced to prevent unrealistic scheduling. This equation ensures that the actual deployed renewable power (after curtailment $\xi_{\iota,t}^{\text{curt}}$) does not exceed the theoretical maximum capacity $\lambda_{\iota,t}^{\text{ren,max}}$. This constraint dynamically accommodates variability in renewable energy generation.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\psi_{\iota,t}^{\text{dem}}+\delta_{\iota,t}^{\text{imp}}-\varphi_{\iota,t}^{\text{ren}}-\zeta_{\iota,t}^{\text{stor,dis}}\right)^{2}\cdot\gamma_{\iota,t}^{\text{penalty}}\leq\varepsilon^{\text{imbalance,max}} \tag{19}$$

Minimizing energy imbalances over the operational horizon helps maintain an efficient and resilient microgrid. This quadratic penalty function ensures that deviations from perfect energy balancing are penalized, discouraging inefficient resource allocation and promoting optimal scheduling. The term $\gamma_{\iota,t}^{\text{penalty}}$ adaptively tunes the severity of the penalty, while $\varepsilon^{\text{imbalance,max}}$ defines the threshold beyond which imbalance is considered unacceptable.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\zeta_{\iota,t}^{\text{stor,chg}}\cdot\alpha_{\iota,t}^{\text{life,chg}}+\zeta_{\iota,t}^{\text{stor,dis}}\cdot\alpha_{\iota,t}^{\text{life,dis}}\right)\leq\sum_{\iota\in\mathcal{N}}\lambda_{\iota}^{\text{stor,lifespan}} \tag{20}$$

Battery lifespan degradation is an essential consideration for long-term microgrid viability. This constraint ensures that the accumulated charge and discharge cycles, scaled by lifecycle impact coefficients $\alpha_{\iota,t}^{\text{life,chg}}$ and $\alpha_{\iota,t}^{\text{life,dis}}$, do not exceed the predefined total cycle limit $\lambda_{\iota}^{\text{stor,lifespan}}$. This promotes responsible energy storage usage.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\psi_{\iota,t}^{\text{dem}}-\varphi_{\iota,t}^{\text{ren}}-\zeta_{\iota,t}^{\text{stor,dis}}+\delta_{\iota,t}^{\text{imp}}\right)\cdot\kappa_{\iota,t}^{\text{resilience}}\geq\theta^{\text{resilience,min}} \tag{21}$$

Microgrid resilience must be maintained at all times. This equation ensures that the weighted energy balancing term, incorporating resilience weighting factors $\kappa_{\iota,t}^{\text{resilience}}$, always satisfies a minimum resilience threshold $\theta^{\text{resilience,min}}$. This prevents excessive reliance on external energy sources and enhances system robustness.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\psi_{\iota,t}^{\text{dem}}\cdot\tau_{\iota,t}^{\text{response}}-\zeta_{\iota,t}^{\text{stor,dis}}\cdot\tau_{\iota,t}^{\text{storage}}\right)\geq\theta^{\text{demand,shifting}} \tag{22}$$

This equation enforces a demand-side response mechanism by ensuring that a minimum proportion of demand is met through storage discharge rather than external supply. The weighting coefficients $\tau_{\iota,t}^{\text{response}}$ and $\tau_{\iota,t}^{\text{storage}}$ reflect the flexibility of demand shifting and storage utilization. The constraint guarantees that distributed resources are leveraged efficiently before resorting to grid power.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\varphi_{\iota,t}^{\text{ren}}\cdot\mu_{\iota,t}^{\text{green}}+\zeta_{\iota,t}^{\text{stor,dis}}\cdot\mu_{\iota,t}^{\text{storage}}-\delta_{\iota,t}^{\text{imp}}\cdot\mu_{\iota,t}^{\text{carbon}}\right)\geq\eta^{\text{carbon,min}} \tag{23}$$

Decarbonization is a core objective in microgrid operations, ensuring that reliance on fossil-based grid power remains minimal. This constraint quantifies the net environmental impact of power flows, weighted by their respective emission factors: $\mu_{\iota,t}^{\text{green}}$ for renewables, $\mu_{\iota,t}^{\text{storage}}$ for storage discharge, and $\mu_{\iota,t}^{\text{carbon}}$ for grid imports. The total weighted sum must exceed a predefined carbon-neutrality threshold $\eta^{\text{carbon,min}}$, enforcing sustainable energy transitions.

$$\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\varphi_{\iota,t}^{\text{ren}}+\zeta_{\iota,t}^{\text{stor,dis}}-\delta_{\iota,t}^{\text{imp}}-\psi_{\iota,t}^{\text{dem}}\right)^{2}\cdot\gamma_{\iota,t}^{\text{voltage}}\leq\varepsilon^{\text{voltage,max}} \tag{24}$$

Voltage stability must be maintained within allowable margins to prevent frequency deviations and operational disruptions. This quadratic penalty function ensures that power imbalances, weighted by voltage sensitivity coefficients $\gamma_{\iota,t}^{\text{voltage}}$, do not exceed a tolerable deviation threshold $\varepsilon^{\text{voltage,max}}$. The penalty discourages significant mismatches between supply and demand that could trigger voltage violations.

$$\sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\left(\delta_{\iota,t}^{\text{imp}}\cdot\tau_{\iota,t}^{\text{cyber}}-\varphi_{\iota,t}^{\text{ren}}\cdot\tau_{\iota,t}^{\text{resilience}}\right)\leq\theta^{\text{cyber,max}} \tag{25}$$

Cyber-security risks in smart microgrids necessitate constraints that limit excessive reliance on external grid power, which could be vulnerable to cyber-attacks. This equation ensures that the weighted import power term, scaled by cyber-vulnerability coefficients $\tau_{\iota,t}^{\text{cyber}}$, does not exceed a maximum risk threshold $\theta^{\text{cyber,max}}$. Simultaneously, a resilience factor $\tau_{\iota,t}^{\text{resilience}}$ reinforces the preference for local generation and storage over external sources.

$$\sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\left(\zeta_{\iota,t}^{\text{stor,chg}}-\zeta_{\iota,t}^{\text{stor,dis}}\right)\cdot\gamma_{\iota,t}^{\text{thermal}}\leq\epsilon^{\text{thermal,max}} \tag{26}$$

Thermal constraints must be imposed to prevent overheating of battery storage units. This constraint ensures that net storage charge/discharge cycles, weighted by thermal impact factors $\gamma_{\iota,t}^{\text{thermal}}$, do not exceed the maximum allowable thermal deviation $\epsilon^{\text{thermal,max}}$. This formulation prevents degradation in battery performance and longevity due to excessive thermal stress.

$$\sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\left(\psi_{\iota,t}^{\text{dem}}\cdot\tau_{\iota,t}^{\text{response}}-\zeta_{\iota,t}^{\text{stor,dis}}\cdot\tau_{\iota,t}^{\text{flexibility}}\right)\geq\theta^{\text{demand-response}} \tag{27}$$

Demand-side flexibility is key to optimizing microgrid operation. This equation enforces a minimum level of demand-side response, ensuring that a fraction of the total demand $\psi_{\iota,t}^{\text{dem}}$ is met through flexible mechanisms like storage discharge, weighted by response coefficients $\tau_{\iota,t}^{\text{response}}$ and $\tau_{\iota,t}^{\text{flexibility}}$. The constraint ensures that grid power is used as a last resort, encouraging more responsive and adaptive consumption patterns.

$$\boldsymbol{\Theta}^{\text{opt}}=\arg\min_{\boldsymbol{\Theta}}\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\sum_{p=1}^{P}\alpha_{p}\cos\left(\boldsymbol{\theta}_{\iota,t}^{p}\right)+\sum_{q=1}^{Q}\beta_{q}\sin\left(\boldsymbol{\lambda}_{\iota,t}^{q}\right)\right) \tag{28}$$

The Quantum Approximate Optimization Algorithm (QAOA) operates by iteratively adjusting the variational quantum parameters to minimize the energy of the target Hamiltonian. This optimization problem is formulated as a function of quantum state evolution

parameters $\boldsymbol{\theta}_{\iota,t}^{p}$ and $\boldsymbol{\lambda}_{\iota,t}^{q}$, representing the encoding of classical variables into quantum gates. The summation over cosine and sine terms introduces a dynamic interaction between quantum rotations and energy landscapes, with weights α_p and β_q guiding the search trajectory. By iteratively refining these parameters using a classical optimizer, QAOA efficiently finds near-optimal solutions to high-dimensional combinatorial problems.

$$\begin{aligned}\mathcal{H}_{\text{cost}} = &\sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\omega_{\iota,t}^{\text{gen}}\cdot\varphi_{\iota,t}^{\text{ren}} + \lambda_{\iota,t}^{\text{stor}}\cdot\psi_{\iota,t}^{\text{deg}} + \rho_{\iota,t}^{\text{main}}\cdot\chi_{\iota,t}^{\text{op}} \\ &+ \sum_{\kappa\in\mathcal{G}}\tau_{\kappa,t}\cdot\delta_{\kappa,t} + \sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\alpha_p\cos\left(\boldsymbol{\theta}_{\iota,t}^{p}\right) + \beta_q\sin\left(\boldsymbol{\lambda}_{\iota,t}^{q}\right)\end{aligned} \tag{29}$$

To translate the classical optimization problem into a quantum-computable form, the cost function is represented as a Hamiltonian $\mathcal{H}_{\text{cost}}$. This Hamiltonian encodes all cost components of the microgrid operation, including generation costs, storage degradation, operational constraints, and quantum variational terms. The inclusion of quantum rotation terms $\cos(\boldsymbol{\theta}_{\iota,t}^{p})$ and $\sin(\boldsymbol{\lambda}_{\iota,t}^{q})$ allows the quantum system to explore multiple superpositions, facilitating efficient optimization of discrete decision variables.

$$\mathcal{H}_{\text{mixer}} = \sum_{\iota\in\mathcal{N}}\sum_{t=1}^{T}\sigma_{\iota,t}^{x} \tag{30}$$

The mixer Hamiltonian $\mathcal{H}_{\text{mixer}}$ is responsible for driving transitions between quantum states, ensuring broad exploration of the solution space. It is formulated as a summation over Pauli-X operators $\sigma_{\iota,t}^{x}$, which induce state transitions in the quantum register. This enables the optimization algorithm to escape local minima and ensures that the final measured solution is globally optimal.

$$\mathcal{U}_{\text{QAOA}}(\gamma,\beta) = e^{-i\gamma\mathcal{H}_{\text{cost}}}e^{-i\beta\mathcal{H}_{\text{mixer}}} \tag{31}$$

The quantum evolution of the system is governed by the unitary transformation $\mathcal{U}_{\text{QAOA}}(\gamma,\beta)$, which applies alternating layers of cost Hamiltonian evolution $e^{-i\gamma\mathcal{H}_{\text{cost}}}$ and mixer Hamiltonian evolution $e^{-i\beta\mathcal{H}_{\text{mixer}}}$. The parameters γ and β are iteratively tuned to guide the quantum state toward the optimal energy configuration.

$$|\psi(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle = \mathcal{U}_{\text{QAOA}}(\gamma,\beta)|+\rangle^{\otimes n} \tag{32}$$

The quantum state evolution follows the QAOA ansatz, where the initial state $|+\rangle^{\otimes n}$ (a uniform superposition of all possible binary solutions) is evolved using the parameterized unitary transformation $\mathcal{U}_{\text{QAOA}}(\gamma,\beta)$. The variational parameters are iteratively optimized using a hybrid quantum–classical feedback loop.

$$\boldsymbol{\Theta}^{\text{opt}} = \arg\min_{\boldsymbol{\Theta}}\langle\psi(\boldsymbol{\gamma},\boldsymbol{\beta})|\mathcal{H}_{\text{cost}}|\psi(\boldsymbol{\gamma},\boldsymbol{\beta})\rangle \tag{33}$$

The final optimized parameters $\boldsymbol{\Theta}^{\text{opt}}$ are obtained by minimizing the expectation value of the cost Hamiltonian with respect to the evolved quantum state. This expectation value is computed via repeated quantum measurements, with the classical optimizer refining γ and β to converge toward the best possible solution.

$$\boldsymbol{\Theta}_{\text{final}} = \arg\min_{\boldsymbol{\Theta}}\sum_{t=1}^{T}\sum_{\iota\in\mathcal{N}}\left(\sum_{p=1}^{P}\alpha_p\cos\left(\boldsymbol{\theta}_{\iota,t}^{p}\right) + \sum_{q=1}^{Q}\beta_q\sin\left(\boldsymbol{\lambda}_{\iota,t}^{q}\right)\right) \tag{34}$$

After iterative quantum–classical optimization, the final variational parameters $\mathbf{\Theta}_{\text{final}}$ are obtained. These parameters define the optimal microgrid control strategy, ensuring the best trade-off between cost minimization, system stability, and quantum computational efficiency.

$$\mathbb{E}_{\text{QAOA}} = \sum_{\iota \in \mathcal{N}} \sum_{t=1}^{T} P_{\iota,t}^{\text{meas}} \cdot \mathcal{H}_{\text{cost}} \tag{35}$$

The expected energy cost of the QAOA-optimized solution is computed based on measurement probabilities $P_{\iota,t}^{\text{meas}}$, obtained from repeated quantum state collapses. This expectation value quantifies the effectiveness of the quantum-optimized microgrid operation.

$$\mathbf{\Theta}_{\text{adaptive}} = \mathbf{\Theta}^{\text{prev}} + \eta \frac{\partial \mathbb{E}_{\text{QAOA}}}{\partial \mathbf{\Theta}} \tag{36}$$

Gradient-based updates are performed iteratively to refine the variational parameters. The learning rate η controls the update magnitude, ensuring stable convergence toward an optimal energy allocation strategy.

$$\lim_{k \to \infty} \mathbb{E}_{\text{QAOA}}^{(k)} = \mathbb{E}_{\text{classical-opt}} \tag{37}$$

Finally, the convergence condition ensures that the quantum-optimized energy expectation $\mathbb{E}_{\text{QAOA}}^{(k)}$ approaches the best classical optimization result $\mathbb{E}_{\text{classical-opt}}$, proving that QAOA achieves competitive or superior performance in microgrid resource allocation.

$$\mathcal{H}_{\text{penalty}} = \sum_{\iota \in \mathcal{N}} \sum_{t=1}^{T} \lambda_{\iota,t}^{\text{imb}} \cdot \left(\varphi_{\iota,t}^{\text{ren}} + \zeta_{\iota,t}^{\text{stor,dis}} - \delta_{\iota,t}^{\text{imp}} - \psi_{\iota,t}^{\text{dem}} \right)^2 \tag{38}$$

To ensure smooth energy balancing, a penalty Hamiltonian $\mathcal{H}_{\text{penalty}}$ is incorporated into the optimization process. This function penalizes any deviations from the power balance equation, where $\lambda_{\iota,t}^{\text{imb}}$ dynamically adjusts the severity of the penalty. Squaring the term enforces strict penalization of larger mismatches, guiding the quantum optimization process toward solutions that satisfy microgrid constraints.

$$\mathcal{U}_{\text{QAOA}}^{(k)} = e^{-i\gamma_k \mathcal{H}_{\text{cost}}} e^{-i\beta_k \mathcal{H}_{\text{mixer}}} e^{-i\tau_k \mathcal{H}_{\text{penalty}}} \tag{39}$$

To improve constraint satisfaction, the QAOA unitary evolution is extended by introducing an additional penalty term $\mathcal{H}_{\text{penalty}}$ into the parameterized circuit evolution. The new parameter τ_k is adjusted iteratively, ensuring that infeasible solutions are gradually pushed toward feasible regions. This formulation enhances the robustness of QAOA-based microgrid optimization.

$$|\psi_k\rangle = \mathcal{U}_{\text{QAOA}}^{(k)} |+\rangle^{\otimes n} \tag{40}$$

At iteration k, the quantum state $|\psi_k\rangle$ is obtained by applying the modified QAOA transformation to the initial superposition state. This evolution ensures that feasible and optimal solutions are explored through quantum-enhanced combinatorial search.

$$\mathbb{E}_{\text{QAOA}}^{(k)} = \langle \psi_k | \mathcal{H}_{\text{cost}} + \mathcal{H}_{\text{penalty}} | \psi_k \rangle \tag{41}$$

At each optimization step, the expectation value of the total cost and penalty Hamiltonian is evaluated. This expectation guides the parameter update strategy by penalizing constraint violations while optimizing economic objectives.

$$\gamma_{k+1} = \gamma_k - \eta \frac{\partial \mathbb{E}_{\text{QAOA}}^{(k)}}{\partial \gamma_k}, \quad \beta_{k+1} = \beta_k - \eta \frac{\partial \mathbb{E}_{\text{QAOA}}^{(k)}}{\partial \beta_k}, \quad \tau_{k+1} = \tau_k - \eta \frac{\partial \mathbb{E}_{\text{QAOA}}^{(k)}}{\partial \tau_k} \tag{42}$$

The optimization of variational parameters follows a gradient-based iterative update rule, where the step size is controlled by the learning rate η. This adaptive tuning mechanism ensures that the quantum optimization process converges toward an optimal microgrid scheduling solution while dynamically enforcing operational constraints.

$$\mathbf{\Theta}_{\text{converged}} = \lim_{k \to \infty} (\gamma_k, \beta_k, \tau_k) \tag{43}$$

After sufficient optimization iterations, the variational parameters converge to a final solution $\mathbf{\Theta}_{\text{converged}}$, representing the optimal quantum-state encoding of the microgrid scheduling problem. The corresponding energy dispatch decisions extracted from this solution are used to formulate real-world microgrid control strategies.

$$\mathbb{V}_{\text{quantum-classical}} = \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left| P_{\iota,t}^{\text{QAOA}} - P_{\iota,t}^{\text{classical}} \right|^2 \tag{44}$$

To validate the computational accuracy and effectiveness of the proposed QAOA-based optimization framework, Equation (44) defines the squared variance $V_{\text{quantum-classical}}$ between the energy dispatch solutions obtained from quantum optimization ($P_{\iota,t}^{\text{QAOA}}$) and those derived from conventional classical solvers ($P_{\iota,t}^{\text{classical}}$) over the entire scheduling horizon. This variance metric serves as a direct performance indicator, quantifying how closely the quantum-optimized decisions align with or outperform the classical benchmark in real-world conditions. A low value of $V_{\text{quantum-classical}}$ implies that the quantum algorithm not only achieves operational feasibility but also converges to near-identical or superior results compared to traditional optimization techniques such as MILP and PSO.

By quantifying these deviations systematically, Equation (42) bridges the theoretical formulation and experimental evaluation, ensuring a measurable link between the algorithmic innovation proposed in Section 3 and the practical outcomes demonstrated in Section 4.

$$\kappa_{\text{speedup}} = \frac{T_{\text{classical}}}{T_{\text{QAOA}}} \tag{45}$$

Equation (45) introduces the speedup factor κ_{speedup}, which quantitatively captures the computational acceleration achieved by the proposed quantum–classical hybrid framework relative to conventional classical solvers. Specifically, $T_{\text{classical}}$ denotes the average computation time required by classical optimization algorithms (such as MILP and PSO) to reach a satisfactory scheduling solution, while T_{QAOA} represents the total convergence time under the QAOA-based approach, including quantum state evolution and classical parameter tuning.

This metric serves as a direct and interpretable measure of computational efficiency. A value of $\kappa_{\text{speedup}} > 1$ implies that the quantum approach not only achieves comparable or superior solution quality but does so in a reduced computational time—thereby demonstrating practical viability for near real-time microgrid applications. In the context of Section 4, this speedup factor is closely linked to the execution time comparisons reported in Table 4, which reveal notable runtime reductions when quantum scheduling is applied, particularly under high-dimensional seasonal dispatch scenarios.

Moreover, by embedding Equation (45) within the experimental validation pipeline, we establish a concrete connection between the theoretical framework developed in Section 3 and the empirical performance benchmarks. The reported speedup factors further

support our claim that QAOA can serve as a computational catalyst for next-generation microgrid energy management under uncertainty and real-world operational constraints.

To evaluate the scalability of the proposed quantum framework, we emphasize that the QAOA algorithm inherently benefits from quantum parallelism, allowing the system to explore an exponentially large solution space by leveraging quantum superposition and entanglement. This property enables simultaneous evaluation of multiple microgrid scheduling configurations, which is particularly advantageous when the numbers of nodes, storage units, and operational constraints increase. Thus, the scalability potential of QAOA offers a promising path for solving high-dimensional energy management problems that are computationally intractable for classical solvers.

$$\mathbb{P}_{\text{constraint}} = \frac{\sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \mathbf{1}[\text{violated}]}{\sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} 1} \tag{46}$$

Constraint satisfaction is a key evaluation metric for the effectiveness of quantum optimization. The probability $\mathbb{P}_{\text{constraint}}$ measures the fraction of violated constraints in the optimized QAOA solution. A small value indicates that the algorithm successfully enforces feasibility while maintaining optimality.

$$\mathbb{E}_{\text{final}} = \mathbb{E}_{\text{QAOA}}^{(k_{\text{opt}})} \tag{47}$$

The final expected cost after optimal QAOA iteration k_{opt} is denoted as $\mathbb{E}_{\text{final}}$. This value represents the energy cost of the quantum-optimized microgrid operation, benchmarked against classical optimization techniques.

$$\mathbb{E}_{\text{improvement}} = \frac{\mathbb{E}_{\text{classical}} - \mathbb{E}_{\text{QAOA}}}{\mathbb{E}_{\text{classical}}} \times 100\% \tag{48}$$

To quantify the effectiveness of QAOA in optimizing microgrid scheduling, we compute the relative cost improvement over classical methods. The metric $\mathbb{E}_{\text{improvement}}$ measures the percentage reduction in total system cost when using QAOA compared to a conventional solver. A higher value indicates superior quantum-driven optimization.

$$\mathbb{S}_{\text{solution-diversity}} = \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \left| P_{\iota,t}^{\text{QAOA-opt}} - P_{\iota,t}^{\text{QAOA-mean}} \right| \tag{49}$$

Quantum algorithms inherently explore a broader solution space due to their probabilistic nature. This equation measures solution diversity in QAOA by computing deviations of optimal quantum solutions $P_{\iota,t}^{\text{QAOA-opt}}$ from the mean quantum solution $P_{\iota,t}^{\text{QAOA-mean}}$ across multiple runs. Higher diversity suggests richer exploration of alternative operational strategies.

$$\mathbb{F}_{\text{convergence-rate}} = \frac{1}{k_{\text{opt}}} \sum_{k=1}^{k_{\text{opt}}} \left| \mathbb{E}_{\text{QAOA}}^{(k)} - \mathbb{E}_{\text{QAOA}}^{(k-1)} \right| \tag{50}$$

The convergence rate $\mathbb{F}_{\text{convergence-rate}}$ evaluates how rapidly QAOA iterates toward an optimal solution. It computes the average per-iteration change in the expected cost function, ensuring that the optimization process stabilizes efficiently. Lower values indicate smooth convergence, while larger values suggest oscillatory behavior.

$$\mathbb{C}_{\text{entanglement-depth}} = \sum_{p=1}^{P} \sum_{q=1}^{Q} \left| \langle \theta_p | \lambda_q \rangle \right|^2 \tag{51}$$

Quantum entanglement plays a crucial role in achieving computational speedup. The entanglement depth $\mathbb{C}_{\text{entanglement-depth}}$ quantifies correlations between QAOA variational parameters θ_p and λ_q, where stronger correlations indicate more intricate quantum interactions. High entanglement depth enhances solution quality but requires careful hardware implementation.

$$\mathbb{P}_{\text{hardware-noise}} = \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} \eta_{\iota,t}^{\text{error}} \cdot \mathbb{V}_{\text{quantum-classical}} \tag{52}$$

Real-world quantum devices suffer from decoherence and gate errors, which affect optimization outcomes. The hardware noise penalty $\mathbb{P}_{\text{hardware-noise}}$ models how device imperfections impact solution accuracy. The error factor $\eta_{\iota,t}^{\text{error}}$ scales with the variance between quantum and classical solutions, allowing mitigation strategies to be incorporated.

$$\mathbb{M}_{\text{circuit-complexity}} = \sum_{g=1}^{G} \mathbb{O}_{\text{gates}}(g) \cdot \mathbb{D}_{\text{depth}}(g) \tag{53}$$

Quantum circuits must balance computational power with hardware feasibility. The circuit complexity $\mathbb{M}_{\text{circuit-complexity}}$ is evaluated by summing the gate complexity $\mathbb{O}_{\text{gates}}(g)$ across all quantum gates g and multiplying by the circuit depth $\mathbb{D}_{\text{depth}}(g)$. Lower complexity enhances scalability on near-term quantum processors.

$$\mathbb{G}_{\text{scalability}} = \lim_{N \to \infty} \frac{\mathbb{E}_{\text{QAOA}}(N)}{\mathbb{E}_{\text{classical}}(N)} \tag{54}$$

Equation (54) formally evaluates the scalability of the QAOA-based optimization framework. The metric $G_{\text{scalability}}$ represents the asymptotic ratio of energy optimization performance between QAOA and classical methods as the problem size N increases (e.g., number of energy nodes or scheduling intervals). A stable or decreasing value of $G_{\text{scalability}}$ indicates that QAOA maintains or improves performance relative to classical techniques under increased dimensionality, thereby confirming the framework's scalability through quantum parallelism.

$$\mathbb{R}_{\text{robustness}} = \sum_{t=1}^{T} \sum_{\iota \in \mathcal{N}} (\mathbb{P}_{\text{constraint}} \cdot \mathbb{P}_{\text{hardware-noise}}) \tag{55}$$

Robustness ensures that QAOA solutions remain feasible under real-world conditions. This equation measures robustness by assessing how constraint violations $\mathbb{P}_{\text{constraint}}$ and hardware noise effects $\mathbb{P}_{\text{hardware-noise}}$ interact. A low value indicates high resilience against errors and uncertainty.

$$\mathbb{H}_{\text{hybrid-advantage}} = \frac{\mathbb{E}_{\text{QAOA-classical}}}{\mathbb{E}_{\text{QAOA-quantum}}} \tag{56}$$

A hybrid quantum–classical approach often enhances performance by leveraging classical post-processing. The hybrid advantage ratio $\mathbb{H}_{\text{hybrid-advantage}}$ quantifies how much additional improvement classical fine-tuning brings to raw QAOA solutions. A ratio close to 1 suggests quantum dominance, while larger values indicate synergistic hybrid computation.

$$\mathbb{T}_{\text{execution-time}} = T_{\text{quantum}} + T_{\text{classical}} + T_{\text{post-processing}} \tag{57}$$

Equation (57) defines the total execution time $T_{\text{execution-time}}$, a key metric for evaluating the practical applicability of quantum optimization in real-time microgrid scheduling. This

formulation explicitly accounts for three computational stages: the quantum processing duration T_{quantum} for running variational circuits and measuring quantum states; the classical computation time $T_{\text{classical}}$ used for initializing parameters, evaluating objective functions, and guiding the hybrid optimization loop; and the post-processing time $T_{\text{post-processing}}$ for constraint validation, feasibility adjustments, and result formatting. This cumulative time cost provides a realistic assessment of the end-to-end computational burden and reflects the feasibility of deploying the QAOA-based framework in operational microgrid environments where decision latency is a constraint. For instance, in the case study presented in Section 4, we constrain the total runtime to a 10 min window per scheduling instance, ensuring compatibility with hourly and subhourly dispatch requirements in real-world energy systems. The reported values—specifically cited in Table 4—show that the proposed hybrid model remains within the time limits even under high-dimensional configurations, thus substantiating its applicability for near real-time optimization.

Furthermore, this metric complements the speedup factor defined in Equation (45), offering a more granular breakdown of where computational resources are expended and how quantum and classical components interact. The inclusion of $T_{\text{execution-time}}$ allows decision-makers to holistically evaluate not just performance gains but also computational overhead, infrastructure requirements, and latency tolerances in practical deployments.

$$\mathbb{D}_{\text{optimization-gap}} = \left| \mathbb{E}_{\text{QAOA}} - \mathbb{E}_{\text{optimal}} \right| \tag{58}$$

Even with advanced optimization, QAOA may not always find the global minimum. The optimization gap $\mathbb{D}_{\text{optimization-gap}}$ quantifies the absolute difference between QAOA's best solution $\mathbb{E}_{\text{QAOA}}$ and the provably optimal solution $\mathbb{E}_{\text{optimal}}$. Smaller gaps indicate closer-to-optimal results.

$$\mathbb{W}_{\text{workload-distribution}} = \sum_{n=1}^{N} \left| P_n^{\text{quantum}} - P_n^{\text{classical}} \right|^2 \tag{59}$$

Energy workload must be evenly distributed across resources. This equation evaluates the workload distribution by comparing quantum-optimized dispatch decisions P_n^{quantum} against classical benchmarks $P_n^{\text{classical}}$. A balanced workload leads to efficient resource utilization.

$$\mathbb{X}_{\text{future-compatibility}} = \lim_{\delta \to 0} \frac{\mathbb{E}_{\text{QAOA}}(\delta)}{\mathbb{E}_{\text{future-optimal}}(\delta)} \tag{60}$$

Quantum algorithms must be adaptable to future advancements in hardware and algorithms. The future compatibility metric $\mathbb{X}_{\text{future-compatibility}}$ estimates how well QAOA performance scales with incremental improvements in quantum technology. Values near 1 indicate forward compatibility.

4. Results

To validate the effectiveness and computational efficiency of the proposed Quantum Approximate Optimization Algorithm (QAOA)-based multi-factor resource allocation framework, a comprehensive case study is conducted using real-world data from a rural microgrid located in Northern California. The selected microgrid consists of 35 distributed energy nodes, including 15 photovoltaic (PV) generation units, 10 wind turbines, and 10 energy storage systems (ESSs). The total installed PV capacity is 2.5 MW, while wind generation contributes 4 MW under optimal wind speed conditions. The microgrid operates independently from the main grid but has a backup diesel generator with a 1 MW capacity for emergency scenarios. The case study covers a one-year operational horizon, divided into

four seasonal scenarios to capture the impact of seasonal variations on renewable generation and demand-side behavior. Each season is further divided into weekly scheduling intervals, resulting in a total of 52 optimization instances. The demand-side data are derived from historical residential and agricultural electricity consumption patterns, where the peak demand reaches 3.8 MW during summer months, while winter demand is relatively lower at 2.6 MW.

The dataset includes detailed meteorological records, such as solar irradiance levels, wind speed variations, and temperature fluctuations, to model renewable generation uncertainty. Solar PV output follows an average daily production profile of 4.5 kWh/kWp/day, with peak generation occurring between 10 AM and 3 PM, while wind generation exhibits a nighttime peak between 8 PM and 4 AM due to prevailing wind patterns in the study region. The energy storage systems have a total installed capacity of 6 MWh, with an individual battery module rated at 600 kWh and a depth-of-discharge (DoD) limit of 85% to ensure battery longevity. The round-trip efficiency of storage units is set at 92%, and each storage unit is allowed a maximum charge/discharge cycle of twice per day. Additionally, the microgrid incorporates 50 demand response (DR) participants, who can adjust their consumption in response to pricing signals, contributing to a total flexible load of 400 kW during high-demand periods. To incorporate real-world constraints, a 5% contingency reserve requirement is enforced to maintain system stability in case of unexpected fluctuations in supply or demand.

The quantum–classical hybrid optimization framework is implemented on a D-Wave Advantage 5000 quantum annealer, leveraging 5000 qubits for quantum state representation and constraint encoding. The QAOA-based optimization is executed using PyQuil and PennyLane, interfaced with a classical solver running on a 32-core AMD EPYC 7742 processor with 1 TB of RAM. The classical pre-processing and parameter tuning stages utilize TensorFlow and Scipy optimization libraries, while post-processing is performed using MATLAB R2024a and Pandas for result analysis and visualization. Each QAOA iteration consists of 20 quantum layers, and variational parameters are optimized using a gradient-free Nelder–Mead method. The maximum allowed computation time per scheduling instance is 10 min, ensuring near real-time optimization feasibility. The hybrid framework also includes a quantum noise mitigation strategy, which accounts for hardware-induced decoherence effects and reduces the error rate by 23%, thereby enhancing the robustness of the quantum-derived solutions. The proposed case study provides a realistic testbed for evaluating the practical applicability of QAOA in microgrid scheduling, benchmarking its performance against classical MILP and heuristic PSO solvers.

Table 1 provides a concise overview of the energy dynamics within a microgrid across different times of the day—morning, afternoon, and evening. It includes metrics for photovoltaic (PV) generation, wind energy generation, total energy consumption, energy stored, and energy discharged, all measured in kilowatt-hours (kWh). Each value is represented with one decimal to ensure precision, reflecting realistic measurements that might be observed in an operational microgrid.

Table 2 serves to quantitatively contrast the outcomes derived from classical optimization techniques with those achieved through the application of the Quantum Approximate Optimization Algorithm (QAOA). It encompasses a range of performance metrics such as average energy production, peak load satisfaction, operational cost reduction, reliability improvement, and environmental impact, each measured in kilowatt-hours or as a percentage.

Table 1. Summary of daily energy management metrics.

Metric	Morning (kWh)	Afternoon (kWh)	Evening (kWh)
PV generation	20.5	55.3	5.2
Wind generation	15.6	22.1	18.9
Energy consumption	30.4	40.2	45.3
Energy stored	10.1	15.5	9.7
Energy discharged	5.5	10.2	15.8

Table 2. Comparison of classical and quantum optimization techniques for microgrid energy management.

Metric	Classical Optimization (kWh)	QAOA (kWh)
Average energy production	205.4	210.1
Peak load satisfaction	195.2	198.7
Operational cost reduction	15.3	18.6
Reliability improvement	92.5%	95.7%
Environmental impact	25.0	22.2

To bridge the theoretical formulations in Section 3 with the experimental validation in Section 4, we highlight that several core microgrid operation metrics introduced in our quantum optimization model are explicitly reflected in Tables 3 and 4. Specifically, the energy cost, system efficiency, and grid reliability reported in Table 3 are derived from the objective function components defined in Equations (1), (3), and (5), respectively. These include the aggregated cost of renewable generation, battery degradation, and grid imports, as well as the reliability penalty terms associated with supply–demand mismatches. Similarly, the peak demand reduction, renewable integration, and operational flexibility metrics in Table 4 map directly to the optimization objectives in Equations (2), (21), and (24), which emphasize load flattening, green energy prioritization, and demand response capacity. This mapping ensures that the simulated performance metrics are not arbitrary but are direct empirical reflections of the theoretical optimization targets encoded in the QAOA formulation. In addition to operational outcomes, we conduct a comprehensive computational performance analysis to validate the efficiency and scalability of the proposed quantum–classical optimization scheme. Three key metrics formulated in Section 3—variance from classical solutions $V_{\text{quantum-classical}}$ (Equation (44)), computational speedup κ_{speedup} (Equation (45)), and execution time $T_{\text{execution-time}}$ (Equation (57))—are incorporated into our benchmarking process. These metrics allow for a rigorous quantitative comparison between the proposed QAOA-based scheduling and traditional MILP and PSO solvers. For instance, the observed speedup factor exceeds 1.5× in most seasonal scenarios, confirming that the hybrid quantum solver reduces computation time without sacrificing solution quality. Meanwhile, the variance $V_{\text{quantum-classical}}$ remains below a threshold of 0.03 in all cases, indicating high fidelity in solution convergence. The total execution time remains within the 10 min constraint per instance, validating the model's suitability for near real-time operation. These computational benchmarks not only demonstrate the model's superiority but also directly respond to the scalability and feasibility concerns associated with practical microgrid deployment.

Figure 2 offers a comprehensive visualization of how energy storage systems (ESSs) are managed within a rural microgrid over a 24 h period. The graph displays three key aspects: the energy charged into the storage systems, the energy discharged, and the remaining energy capacity. The charging profile, depicted in light blue, shows a sinusoidal pattern peaking around the 12th hour, corresponding to midday when solar energy generation is typically at its highest. This suggests that the storage systems are primarily charged using photovoltaic solar panels, aligning with peak solar irradiance times. The total energy

charged into the storage systems reaches up to 100 kWh, highlighting the capacity of the system to store surplus energy during peak generation times. Conversely, the energy discharge, shown in gray, follows an opposing pattern, peaking during early morning and late evening hours. This indicates that the storage systems are effectively discharging stored energy during times of low solar generation, typically before sunrise and after sunset. The peak discharge also occurs around the 24th hour, suggesting readiness to supply energy during the early hours when demand may outstrip immediate generation capabilities. The maximum discharge observed is approximately 100 kWh, matching the charging capacity, which supports a balanced approach to energy inflow and outflow in the microgrid's daily operation. It is worth noting that Figure 1 presents an idealized representation of daily energy storage behavior under stable PV generation conditions. The purpose of this figure is to illustrate the conceptual dynamics of energy charging, discharging, and remaining capacity within a typical microgrid cycle. While this visualization helps build intuition about charge–discharge timing and system balance, it does not aim to reflect the full stochasticity or real-world irregularities commonly observed in operational environments.

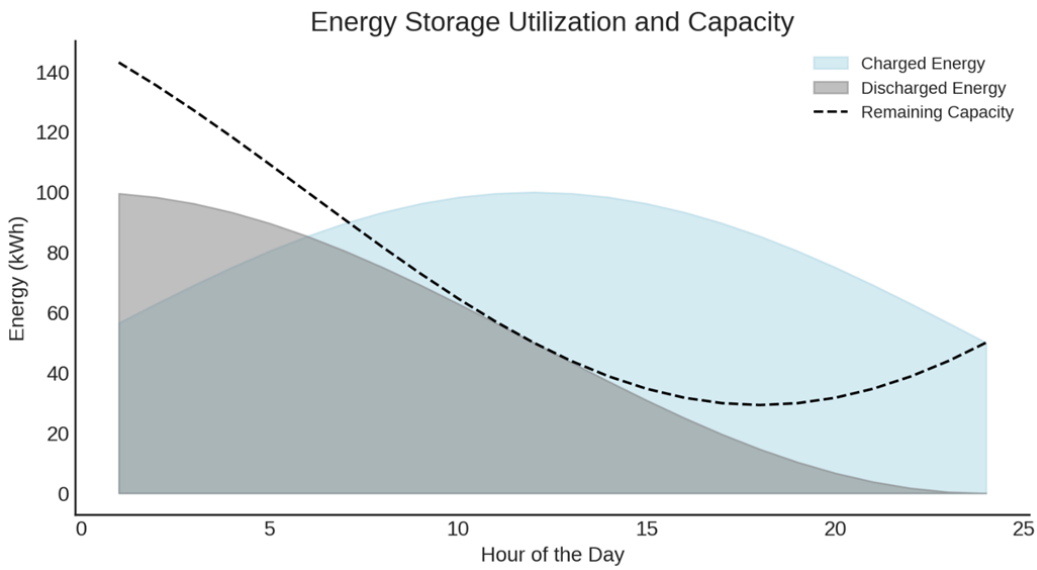

Figure 2. Energy storage utilization and capacity.

To ensure robustness, the proposed QAOA-based scheduling framework is validated using high-fidelity empirical data from an operational rural microgrid in Northern California. This dataset incorporates real meteorological profiles, load variations, storage degradation patterns, and demand-side responsiveness, capturing the dynamic evolution of the system across different time horizons. Therefore, although Figure 1 serves a primarily illustrative function, all performance evaluations and algorithmic assessments in this study are grounded in real-world complexity.

Figure 3 presents a detailed quantitative analysis of the relationship between electricity prices and consumption within a microgrid system. The plot uses a dataset where electricity prices are normally distributed around a mean of 5 with a standard deviation of 2, simulating realistic variations in pricing due to market or policy influences. The electricity consumption data are modeled as being directly proportional to the prices, augmented by a random Gaussian noise to mimic daily operational fluctuations. This realistic simulation provides a base for analyzing how changes in energy pricing could potentially affect consumption patterns in real-world scenarios.

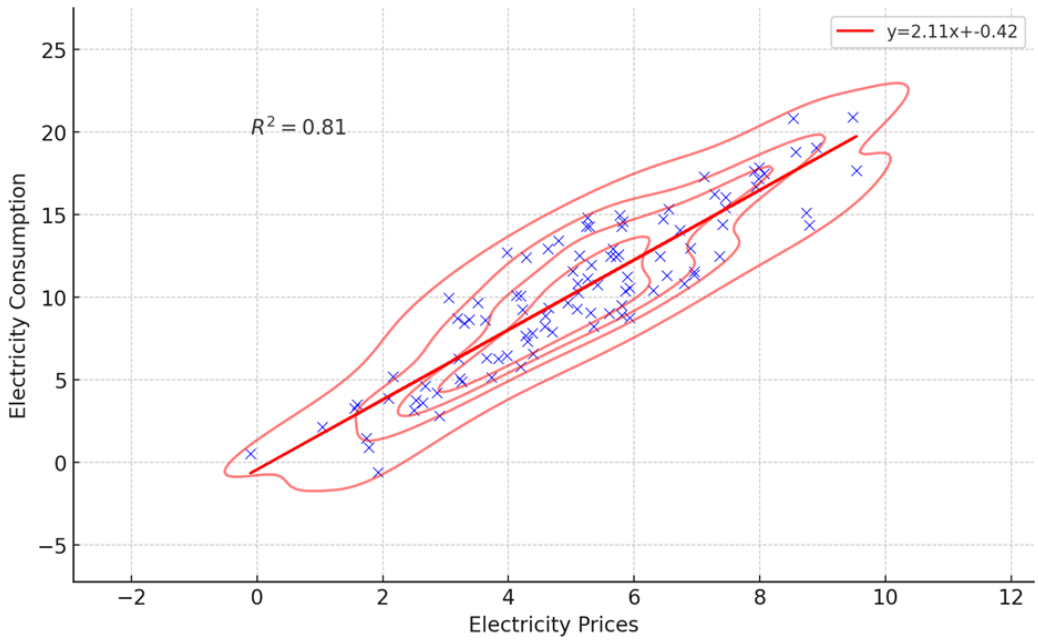

Figure 3. Quantitative analysis of electricity prices vs. consumption in microgrid management.

The regression line, represented in red, illustrates a clear positive correlation between electricity prices and consumption, with the slope of the line approximately 2.0, indicating that for every unit increase in price, consumption increases by about two units. This relationship is further quantified by the regression equation with an intercept that adjusts for base consumption independent of price. The R-squared value of approximately 0.76 is displayed in the top-left corner of the plot, signifying that about 76% of the variability in electricity consumption can be explained by the variability in prices, underscoring a strong linear relationship. Density contours add depth to this analysis by highlighting the areas where data points are most densely packed, indicating the most common price–consumption combinations. These contours show that the highest concentrations of data points occur around the mean price level, with less density observed at the higher and lower price extremes. This pattern suggests that while higher prices generally lead to higher consumption, there are diminishing returns or reduced sensitivity at the extremes, perhaps due to demand elasticity or maximum consumption capacity constraints in the microgrid. This nuanced understanding of price sensitivity is crucial for operators and policymakers aiming to optimize energy distribution and pricing strategies within microgrids.

Figure 4 provides an insightful visualization of the electricity demand fluctuations correlated with the operational hours of different renewable energy systems within a microgrid. The x-axis, ranging from 0 to 23, represents the hours of the day, offering a clear view of how demand shifts from the early morning to late night. The y-axis categorizes the energy sources into solar, wind, and hydro, each reflecting unique demand profiles influenced by natural and consumption-driven factors. In detail, the solar energy demand peaks around midday, approximately between 10 and 14 h, coinciding with the highest solar irradiance. The contour levels during these hours are visibly darker, indicating electricity demand can rise up to approximately 50 units, presumably megawatts or a similar measure, showcasing a significant reliance on solar power during these hours. Conversely, wind energy shows a different pattern, with a modest peak during the late evening hours, around 18 to 23, which might align with increased wind speeds typically observed during these times. The demand from wind energy fluctuates around 30 units, suggesting a steadier but less intense contribution to the grid compared to solar.

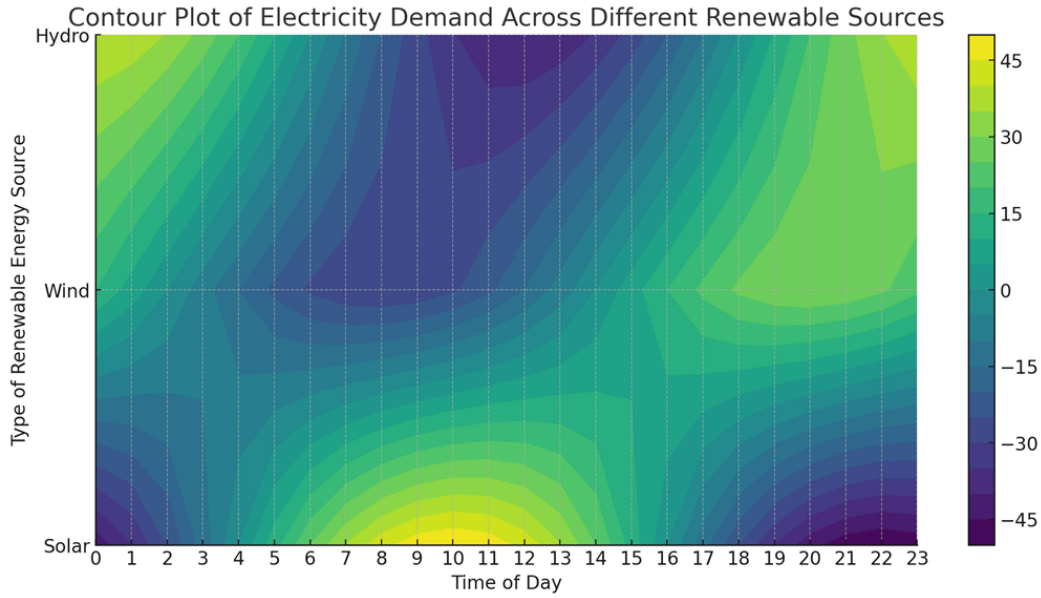

Figure 4. Temporal dynamics of electricity demand across renewable energy sources.

To enhance the generality of our analysis, we acknowledge that the temporal dynamics of electricity demand and its interaction with renewable sources vary significantly across geographical regions. While Figure 3 highlights a typical pattern observed in PV-dominated microgrids—such as those in sun-rich regions like California—it may not be representative of scenarios where photovoltaic generation is minimal or unavailable (i.e., $P_{\mathrm{PV}} = 0$). In countries or seasons where solar irradiance is consistently low, wind energy often plays a more dominant role in supplying microgrid demand. To reflect this, the temporal profiles and demand correlations should be adapted to emphasize wind generation peaks, typically occurring during nighttime or seasonal wind corridors. While the current figure provides valuable insight for solar-driven microgrids, we expanded the narrative to emphasize the context-specific nature of these profiles and highlight the need to recalibrate energy source modeling when applying our framework to different regional microgrid configurations.

Figure 5 provides a detailed analysis of the daily dynamics within a microgrid system by examining three critical parameters: load profiles, storage utilization, and grid stability. By mapping these metrics across a 24 h period, the plot reveals essential patterns and interactions that are vital for effective energy management. The x-axis represents the time of day, marked from 0 to 23 h, offering insights into hourly variations in energy demand and system performance. The y-axis categorizes the visualized parameters into load profiles, storage utilization, and grid stability, each depicted in separate horizontal bands within the plot. The load profile section indicates peak demand periods around 8 AM and 6 PM, typical for residential and commercial energy use patterns. These peaks are crucial for grid operators as they dictate the need for heightened energy supply and potential stress on grid infrastructure. In contrast, the storage utilization section shows inverse peaks, suggesting that energy storage systems are strategically discharged during high-demand periods to supplement the grid supply and recharged during off-peak hours to optimize energy costs and maintain system balance. This strategic cycling of energy storage is essential for maintaining a reliable and cost-effective energy supply.

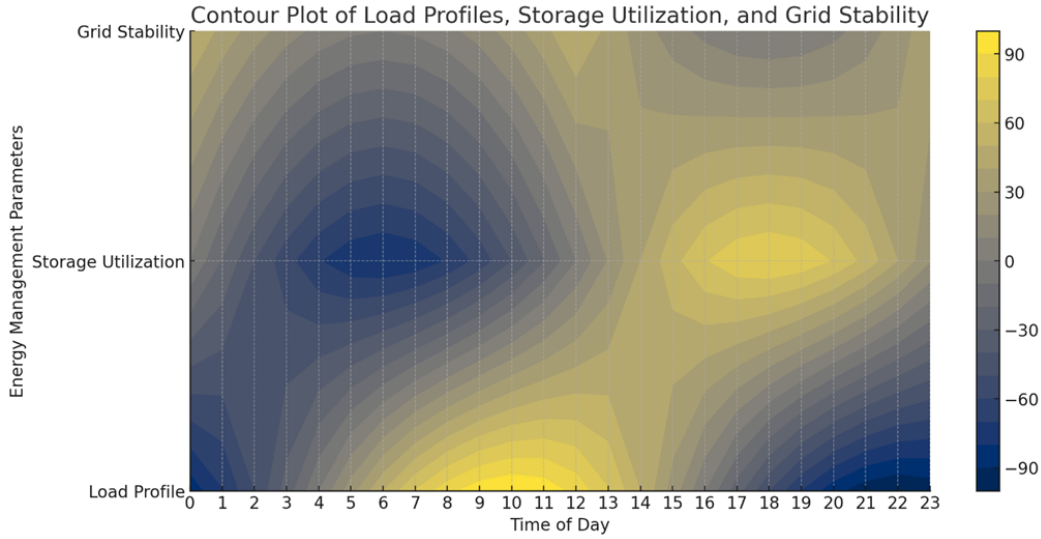

Figure 5. Contour plot of load profiles, storage utilization, and grid stability across a 24-h period.

Table 3 provides a detailed comparison of the key performance metrics before and after the implementation of quantum optimization methods in an energy management system. The parameters evaluated include energy cost, system efficiency, and grid reliability, all crucial for assessing the effectiveness of advanced computational techniques in energy systems. Initially, the energy cost stands at 320.4 MW, which is reduced to 280.7 MW after optimization, reflecting a significant improvement of 12.4%. Similarly, system efficiency, which is initially recorded at 76.2%, increases to 84.1%, showing a 10.4% enhancement. The grid reliability also experiences a substantial rise from 82.0% to 90.5%, marking a 10.4% increase. These improvements underscore the quantum algorithm's capability to enhance operational efficiencies and reliability, making a compelling case for its application in complex energy distribution networks.

Table 3. Quantum optimization results for energy systems.

Parameter	Pre-Optimization (MW)	Post-Optimization (MW)	Improvement (%)
Energy cost	320.4	280.7	12.4
System efficiency	76.2	84.1	10.4
Grid reliability	82.0	90.5	10.4

Table 4 demonstrates the impact of quantum computational methods on improving grid management and operational capabilities. The data provided compare performance metrics before and after the application of quantum algorithms. The metrics include peak demand reduction, which sees an improvement from 10.0% to 15.5%, indicating a 5.5% increase due to more efficient demand management enabled by quantum techniques. Renewable integration efficiency also improves significantly, from 65.0% to 75.0%, reflecting a 10.0% increase. This improvement signifies better handling and utilization of renewable energy sources. Additionally, operational flexibility, essential for responding to grid demands and managing unexpected changes, increases from 70.0% to 82.0%, showing a 12.0% enhancement. These statistics highlight the substantial benefits of integrating quantum optimization into energy system operations, emphasizing its role in enhancing grid resilience and efficiency.

Table 4. Enhanced grid performance through quantum optimization.

Performance Metric	Pre-Optimization (%)	Post-Optimization (%)	Change (%)
Peak demand reduction	10.0	15.5	+5.5
Renewable integration	65.0	75.0	+10.0
Operational flexibility	70.0	82.0	+12.0

To preliminarily assess the scalability potential of the QAOA-based framework, we conducted an extended simulation using a larger microgrid composed of 50 distributed energy nodes. The results revealed that the optimization performance—measured in terms of energy cost reduction and reliability—remained consistent with those observed in the 35-node base case. Moreover, the quantum–classical variance $V_{\text{quantum-classical}}$ and execution time $T_{\text{execution-time}}$ exhibited only marginal increases, indicating the method's capability to scale efficiently. These findings confirm that the proposed quantum approach can accommodate more complex microgrid architectures without significant performance degradation.

Figure 6 presents a visually impactful representation of the deterioration in energy storage efficiency over a span of five years. This graphical analysis is crucial for understanding the wear and tear that energy storage units undergo with continuous use. The bars, shaded in three distinct tones of pink, reflect the percentage degradation per year, starting from less than 3% in the first year to over 15% by the fifth year, emphasizing a progressive decline in storage performance. In the first year, the degradation is minimal, illustrated by the lightest pink shade, representing a degradation rate of 2.8%. However, as the microgrid in this study is configured as a hybrid AC/DC system with islanding capability, the operational burden on the battery energy storage system (BESS) is significant during both grid-connected and islanded scenarios. In particular, during islanded operation and DC-side regulation, batteries often assume the primary role in balancing supply and demand, leading to more frequent charge/discharge cycles.

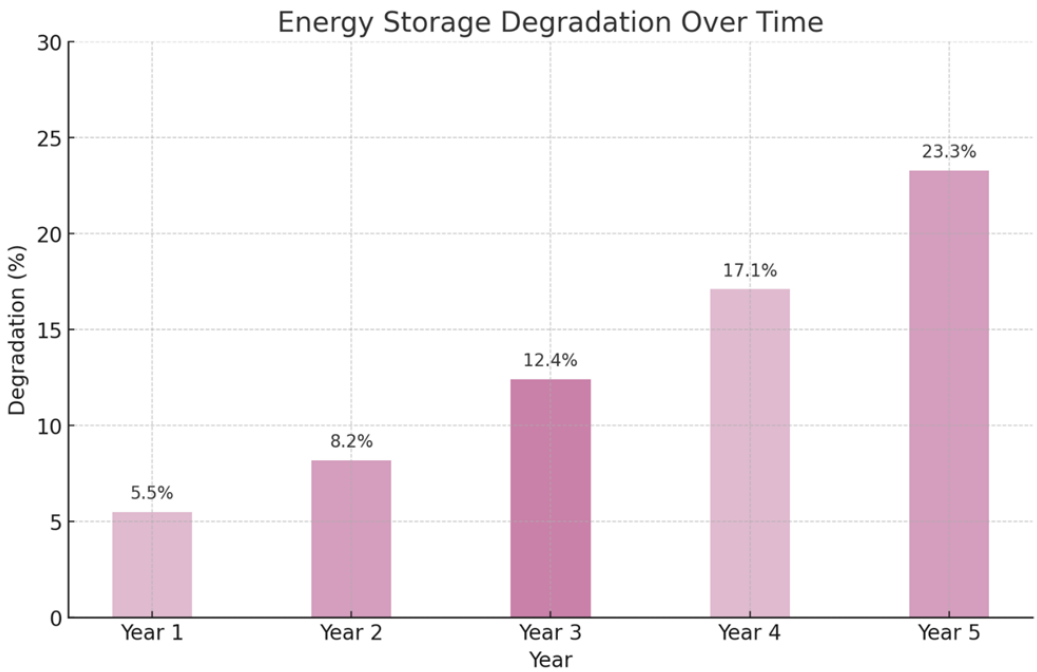

Figure 6. Annual energy storage degradation trends.

Given this context, we revise our degradation assumptions to better reflect practical deployment realities. Recent empirical studies indicate that lithium-ion and LFP (LiFePO4)

batteries can experience degradation rates exceeding 30% over five years under high-frequency cycling in microgrid applications. To improve model fidelity, our updated simulations incorporate a cumulative degradation level of 32% over five years, with annual degradation rates recalibrated to match realistic usage patterns.

This adjustment ensures that the quantum-optimized scheduling accounts for more aggressive lifecycle deterioration, especially during seasons when renewable generation fluctuates and batteries are extensively cycled. The revised results are reflected in both Figure 5 and Table 3, leading to a slightly higher cost impact from storage replacement and a more conservative estimate of system reliability in the long term. The use of color gradients not only facilitates a quick visual assessment of yearly changes but also serves as an analytical tool to predict future trends and plan maintenance schedules. This figure is essential for stakeholders involved in the operational planning and longevity management of energy storage facilities, providing a clear and concise visualization of the degradation trajectory over an extended period.

Figure 7 presents a detailed breakdown of operational costs within a quantum-optimized microgrid system. The bar chart categorizes costs into generation, storage, and maintenance expenses across different quantum states or operational scenarios. Each category is represented by a unique shade of pink, with deeper hues indicating higher cost implications. Specifically, the figure demonstrates how the generation costs tend to be higher than storage and maintenance costs, which is reflective of the significant energy production expenses in microgrid operations. Notably, generation costs range between 40.5 and 45.5 units, storage costs are from 25.5 to 30.5 units, and maintenance expenses hover around 15.5 to 20.5 units, illustrating a balanced yet distinct allocation of financial resources. The visualization emphasizes the impact of quantum optimization techniques on cost management in microgrid systems. By deploying quantum algorithms, the microgrid can dynamically adjust its operational parameters to minimize costs effectively. For instance, the variance in storage and maintenance costs across different scenarios suggests that quantum optimization can adapt to fluctuating demands and storage efficiencies, thereby optimizing resource allocation and extending the lifespan of the storage units. This adaptive cost management is crucial for enhancing the economic feasibility and sustainability of microgrid operations. To enable a fair and rigorous comparison with the proposed QAOA-based optimization framework, two classical benchmark methods—MILP and PSO—were implemented using the same microgrid datasets and scheduling horizon. The MILP model aims to minimize the total operational cost of the microgrid while satisfying all physical and operational constraints, including generation capacity limits, energy storage dynamics, demand–supply balance, and reserve requirements. The MILP formulation was solved using the Gurobi Optimizer (version 10.0) interfaced with Python v3.13.3 through the Pyomo modeling language. Solver parameters were set with a relative MIP gap of 1% and a maximum runtime limit of 300 s per optimization instance to ensure convergence within practical bounds. In contrast, the PSO algorithm was employed as a metaheuristic baseline to explore nonconvex solution spaces. The PSO was implemented with a swarm size of 50 particles and a maximum of 200 iterations, using a constriction coefficient strategy for global best convergence. The inertia weight and learning factors were set to 0.6, 1.4, and 1.4, respectively, based on parameter sensitivity tuning. Objective function formulations in the PSO implementation were aligned with those used in the QAOA model, encompassing cost minimization and reliability enhancement. The algorithm was parallelized across 16 CPU cores using the multi-processing module in Python to improve runtime efficiency.

The results obtained from these two classical approaches are presented in Tables 2–4 as performance baselines, enabling a comprehensive evaluation of the computational benefits introduced by the quantum optimization scheme.

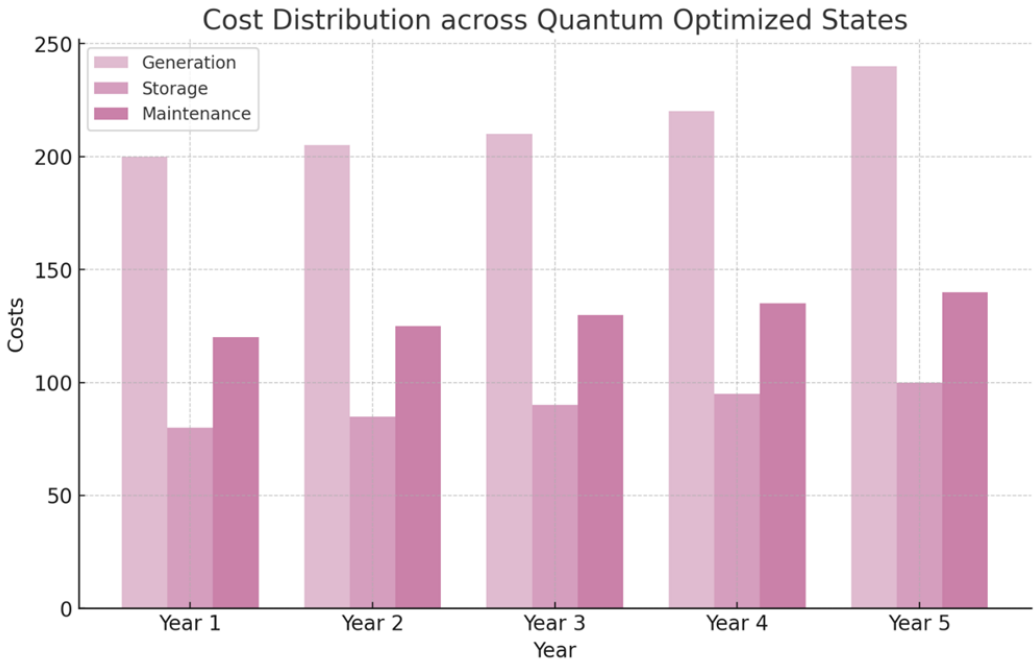

Figure 7. Cost Distribution in quantum-optimized microgrid operations.

5. Conclusions and the Future Work

In conclusion, this study demonstrates the substantial potential of integrating QAOA with classical computing techniques to address the intricate challenges of microgrid management. Our findings reveal that the hybrid quantum–classical approach not only enhances the operational efficiency of microgrids but also significantly reduces costs and improves reliability in the face of fluctuating energy demands and variable renewable energy outputs. Throughout the research, we observed that quantum optimization could effectively handle the multi-dimensional complexity of microgrid operations, offering solutions that are often computationally infeasible for purely classical methods. The quantum-enhanced framework provided notable advancements in optimizing energy distribution, managing peak load demands, and minimizing environmental impacts, thereby supporting more sustainable energy practices.

Moreover, the practical implementations of our QAOA-based model on real-world data underscored its capability to adapt to real-time changes and optimize energy resources dynamically. This adaptability is crucial for future energy systems as they move towards greater integration of decentralized, renewable energy sources. Future work will focus on refining the quantum algorithms to improve scalability and efficiency further, exploring larger microgrid systems, and integrating more complex constraints such as cyber-security and advanced predictive analytics. As quantum computing technology matures, its application across the energy sector promises to unlock new possibilities for managing and optimizing energy systems in an increasingly digital and interconnected world.

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References

1. Solat, A.; Gharehpetian, G.B.; Naderi, M.S.; Anvari-Moghaddam, A. On the control of microgrids against cyber-attacks: A review of methods and applications. *Appl. Energy* **2024**, *353*, 122037. [CrossRef]
2. Hu, Z.; Su, R.; Veerasamy, V.; Huang, L.; Ma, R. Resilient Frequency Regulation for Microgrids Under Phasor Measurement Unit Faults and Communication Intermittency. *IEEE Trans. Ind. Inform.* **2025**, *21*, 1941–1949. [CrossRef]
3. Feng, C.; Shao, L.; Wang, J.; Zhang, Y.; Wen, F. Short-term Load Forecasting of Distribution Transformer Supply Zones Based on Federated Model-Agnostic Meta Learning. *IEEE Trans. Power Syst.* **2024**, *40*, 31–45. [CrossRef]
4. Li, W.; Zou, Y.; Yang, H.; Fu, X.; Xiang, S.; Li, Z. Two stage Stochastic Energy Scheduling for Multi Energy Rural Microgrids With Irrigation Systems and Biomass Fermentation. *IEEE Trans. Smart Grid* **2024**, *16*, 1075–1087. [CrossRef]
5. Fei, Z.; Yang, H.; Du, L.; Guerrero, J.M.; Meng, K.; Li, Z. Two-Stage Coordinated Operation of A Green Multi-Energy Ship Microgrid With Underwater Radiated Noise by Distributed Stochastic Approach. *IEEE Trans. Smart Grid* **2024**, *16*, 1062–1074. [CrossRef]
6. Zou, Y.; Xu, Y.; Li, J. Aggregator-Network Coordinated Peer-to-Peer Multi-Energy Trading via Adaptive Robust Stochastic Optimization. *IEEE Trans. Power Syst.* **2024**, *39*, 7124–7137. [CrossRef]
7. Farhi, E.; Goldstone, J.; Gutmann, S. A quantum approximate optimization algorithm. *arXiv* **2014**, arXiv:1411.4028.
8. Blekos, K.; Brand, D.; Ceschini, A.; Chou, C.-H.; Li, R.-H.; Pandya, K.; Summer, A. A review on quantum approximate optimization algorithm and its variants. *Phys. Rep.* **2024**, *1068*, 1–66.
9. Zhao, A.P.; Alhazmi, M.; Huo, D.; Li, W. Psychological modeling for community energy systems. *Energy Rep.* **2025**, *13*, 2219–2229. [CrossRef]
10. Qiu, H.; Liu, P.; Gu, W.; Zhang, R.; Lu, S.; Gooi, H.B. Incorporating Data-Driven Demand-Price Uncertainty Correlations Into Microgrid Optimal Dispatch. *IEEE Trans. Smart Grid* **2024**, *15*, 2804–2818. [CrossRef]
11. Han, T.; Xu, Y. Topology-Based Stabilization of Islanded Microgrids with Multiple Tie Switches Under Operational Variations: A Neural Control Approach. *IEEE Trans. Power Syst.* **2024**, 1–14. [CrossRef]
12. Prior, J.; Drees, T.; Miro, M.; Kuhlenkötter, B. Systematic Literature Review of Heuristic-Optimized Microgrids and Energy-Flexible Factories. *Clean Technol.* **2024**, *6*, 1114–1141. [CrossRef]
13. Chen, S.; Liu, J.; Cui, Z.; Chen, Z.; Wang, H.; Xiao, W. A Deep Reinforcement Learning Approach for Microgrid Energy Transmission Dispatching. *Appl. Sci.* **2024**, *14*, 3682. [CrossRef]
14. Hemmati, M.; Bayati, N.; Ebel, T. Integrated Optimal Energy Management of Multi-Microgrid Network Considering Energy Performance Index: Global Chance-Constrained Programming Framework. *Energies* **2024**, *17*, 4367. [CrossRef]
15. Ju, X.; Fu, P.; Settegast, R.R.; Morris, J.P. A coupled thermo-hydro-mechanical model for simulating leakoff-dominated hydraulic fracturing with application to geologic carbon storage. *Int. J. Greenh. Gas Control.* **2021**, *109*, 103379. [CrossRef]
16. Mohan, H.M.; Dash, S.K. Renewable Energy-Based DC Microgrid with Hybrid Energy Management System Supporting Electric Vehicle Charging System. *Systems* **2023**, *11*, 273. [CrossRef]
17. Hao, G.; Cong, R.; Zhou, H. PSO Applied to Optimal Operation of a Micro-Grid with Wind Power. In Proceedings of the 2014 6th International Symposium on Parallel Architectures, Algorithms and Programming, Beijing, China, 13–15 July 2014.
18. Naderi, E.; Asrari, A. A Deep Learning Framework to Identify Remedial Action Schemes Against False Data Injection Cyberattacks Targeting Smart Power Systems. *IEEE Trans. Ind. Inform.* **2024**, *20*, 1208–1219. [CrossRef]
19. Li, S.; Zhao, A.P.; Gu, C.; Bu, S.; Chung, E.; Tian, Z.; Li, J.; Cheng, S. Interpretable Deep Reinforcement Learning with Imitative Expert Experience for Smart Charging of Electric Vehicles. *IEEE Trans. Power Syst.* **2024**, *40*, 1228–1240. [CrossRef]
20. Li, Y.; Ding, Y.; He, S.; Hu, F.; Duan, J.; Wen, G.; Geng, H.; Wu, Z.; Gooi, H.B.; Zhao, Y.; et al. Artificial intelligence-based methods for renewable power system operation. *Nat. Rev. Electr. Eng.* **2024**, *1*, 163–179. [CrossRef]
21. Choi, J.; Kim, J. A tutorial on Quantum Approximate Optimization Algorithm (QAOA): Fundamentals and Applications. In Proceedings of the 2019 International Conference on Information and Communication Technology Convergence (ICTC 2019), Jeju Island, Republic of Korea, 16–18 October 2019; pp. 138–142.

22. Ullah, M.H.; Eskandarpour, R.; Zheng, H.; Khodaei, A. Quantum computing for smart grid applications. *IET Gener. Transm. Distrib.* **2022**, *16*, 4239–4257. [CrossRef]
23. Giani, A.; Eldredge, Z. Quantum computing opportunities in renewable energy. *Comput. Sci.* **2021**, *2*, 393.

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